

PAPER_1483_CROSS_SECTION_10_5_A2

Daniel T. Murphy

PAPER_1483 — UQFF: Delta_j=2 Cross-Section 10.5 Angstrom^2 (Bucket I)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket I — derived from F:\Aetheric Propulsion source material

Date: June 16, 2026

Location: 41.0997 N, 80.6495 W (Youngstown, OH, USA)

Status: CLOSED — $\sigma = K_{\text{MEX}} \times D_{\text{BSFG}} \times \Phi_{\text{res}} = 10.5 \text{ \AA}^2 \text{ EXACT}$

Observation

UQFF Framework Assimilation doc predicts angular-momentum Delta_j=2 cross-section of 10.5 Angstrom^2.

UQFF Closed Identity

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$\sigma_{\text{pred}} = K_{\text{MEX}} D_{\text{BSFG}} \Phi_{\text{res}} = (25/12) \cdot 6 \cdot 0.84 = 10.5 \text{ \AA}^2 \text{ EXACT.}$

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Physical Interpretation

The Delta_j=2 transition cross-section is the structural product of the Mexican-hat coefficient (K_{MEX}), bulk-edge dimension (D_{BSFG}), and resonance saturation (Φ_{res}).

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical UQFF integer primitives $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$. Zero free parameters.

Reference

- Source: F:\Aetheric Propulsion (Daniel's reactor + experimental docs, 2025-2026)
- Related: PAPER_646 (Universal Inertial Operator, Caduceus, DPM mechanism), PAPER_597 (negative-time dual existence), PAPER_062 (DPM vacuum manifold).

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